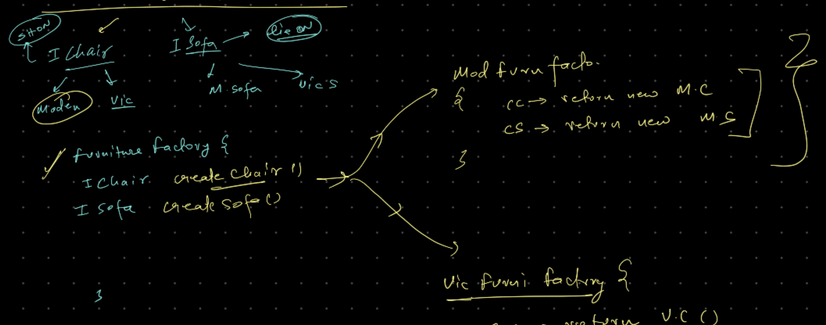
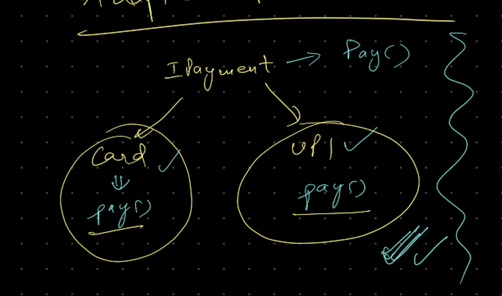
Astract factory Design Pattern



Adapter Pattern




```
Processor {
    IPayment ip;

    process() {
        sout →
        ip.pay();
        sout →
    }
}
```

```
CashPayment {
    makePayment() {
        complicated
    }
}
```

Goal:→ make cashpayment class adaptive to the processor without changing anything in CashPayment.

CashPaymentAdapter → IPayment

```
CashPayment cp;
CashPaymentAdapter(CashPayment cp) cp → cp

pay() {
    cp.makePayment();
}
```

IPayment → pay()

Card → @override pay — C1

UPI @override pay — C2

(3) (X) 6 new classes

uses C → (1) Fee Deduction { (2) Reward { (3) Both Fee & Reward

```
abstract DecoratorBase {
    IPayment ip
    abstract process()
}

FeeDecorator {
    process() {
        fee
        ip.pay
    }
}

RewardDecora {
    process {
        ip.pay
        Reward
    }
}

BothDecora {
    process {
        fee
        ip.pay
        Reward
    }
}
```